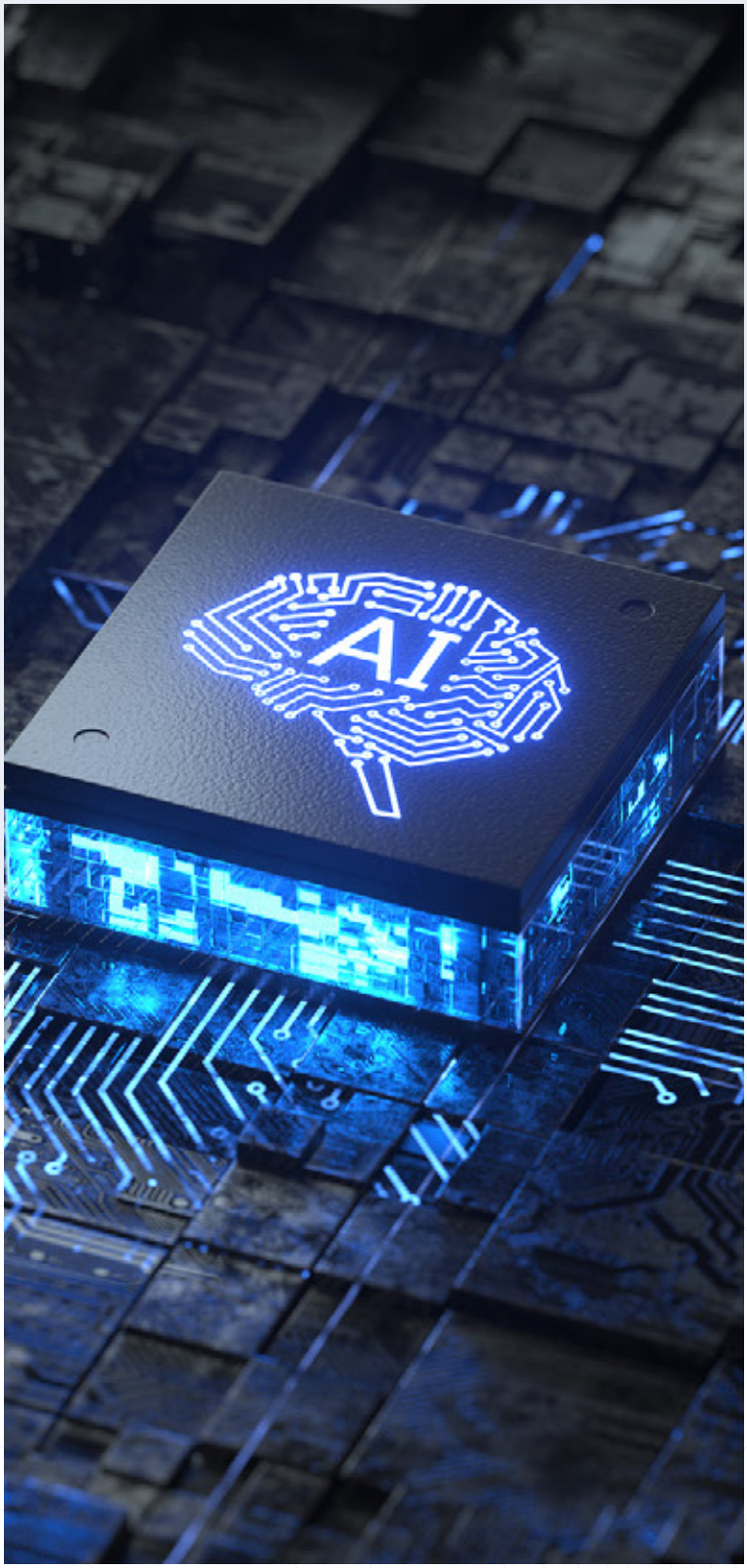
JOHNS HOPKINS
WHITING SCHOOL of ENGINEERING

Great Learning

CERTIFICATE PROGRAM IN APPLIED GENERATIVE AI

16 Weeks | Online | Live Mentored Learning Sessions

GENERATIVE AI: DRIVING INNOVATIVE SOLUTIONS AND GROWTH ACROSS INDUSTRIES



Generative AI has brought a fundamental shift in how we interact with technology and the world around us. From text and image creation to coding, tasks that once required specialized knowledge and hours of effort can now be accomplished with a few simple prompts. Unlike traditional AI, which relies on pre-programmed rules, Generative AI uses patterns and examples from its training data to generate original and diverse outputs, enabling innovative applications across various industries. According to Grand View Research, the global Generative AI market size is expected to grow at a compound annual growth rate (CAGR) of 36.5% from 2024 to 2030. This rapid growth underscores the increasing reliance on GenAI to drive innovation across industries.

Using Generative AI isn't just about understanding the technology—it's much more. To succeed, businesses must align Generative AI with their goals, encourage collaboration, and handle ethical considerations. As Generative AI advances, it provides new opportunities to drive growth and handle the complexities of today's business world.

ABOUT THE PROGRAM

The **Certificate Program in Applied Generative AI** is a transformative 16-week online learning program tailored to equip professionals with advanced skills in Generative AI. The program is led by Dr. Ian McCulloh, who spearheads AI continuing education at Johns Hopkins University and formerly led the US Federal AI practice at Accenture; Dr. Pedro Rodriguez, a faculty member in the Johns Hopkins University Artificial Intelligence program, who oversees work of more than 250 researchers developing innovative AI solutions for government projects, and leads the Information Science branch at the Johns Hopkins Applied Physics Laboratory; and Dr. Iain Cruickshank, who holds faculty positions in Computer Science at both Johns Hopkins University and Carnegie Mellon University, and serves as a Data Scientist and AI Engineer for the U.S. Army.

This program combines theoretical foundations with practical case studies. Learners will explore essential topics such as Large Language Models, Natural Language Processing, and Retrieval Augmented Generation. The curriculum is designed to provide a deep understanding of Generative AI applications, including text, image, and multimedia content generation.

The program offers live mentored learning sessions, personalized support from dedicated Program Managers, and access to peer networks and academic forums. Designed for data and technology professionals, STEM graduates, technology consultants and individuals looking to apply Generative AI in business contexts, it offers hands-on experience in developing and deploying advanced Generative AI solutions. The curriculum also covers ethical considerations and industry best practices, ensuring participants are well-prepared to integrate Generative AI responsibly and effectively into their business strategies.

PROGRAM HIGHLIGHTS

Format: Online (Recorded Video Lectures + Weekly Interactive Mentored Learning)

Expert Guidance: Monthly Live Virtual Masterclasses by JHU Faculty

Duration: 16 Weeks

Weekly Commitment: 8-10 Hours per Week

Real World Case-Based Learning: Hands-on Learning with 10+ Case Studies and 2 Projects

Learning Support: Get Personalized Assistance from a Dedicated Program Manager, and Academic Support Through the GL Community, Project Discussion Forums, and Peer Groups

Recognition: Earn a Certificate From Johns Hopkins University

10 CEUs: Earn 10 Continuing Education Units (CEUs) on Successful Completion of the Program

JOHNS HOPKINS UNIVERSITY ADVANTAGE

Johns Hopkins University (JHU) is a prestigious institution consistently ranked among the top 10 universities in the U.S and has a strong reputation in research and education. It is well-positioned to offer compelling differentiators for its continuing education programs in AI.

WORLD-CLASS FACULTY EXPERTISE

The courses are taught by faculty who are distinguished in academia and have real-world experience leading AI practices at Fortune 500 companies. This blend of academic rigor and industry expertise offers students unique insights into the practical applications of AI.

PRESTIGE AND REPUTATION

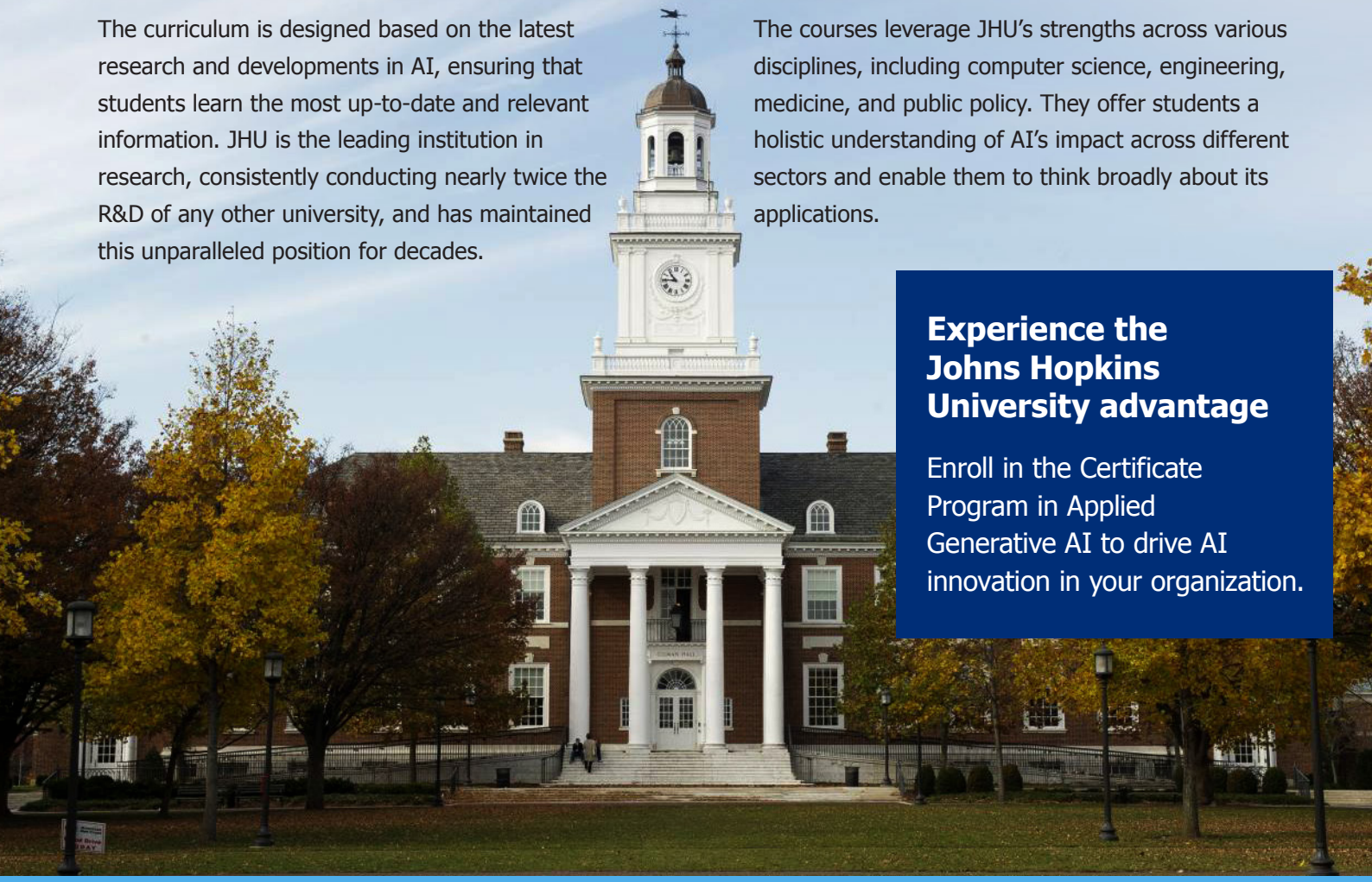
JHU is consistently ranked among the top ten universities globally. Earning a certificate from JHU carries significant weight and can enhance career opportunities for graduates by associating them with a recognized leader in education and research.

RESEARCH-DRIVEN CURRICULUM

The curriculum is designed based on the latest research and developments in AI, ensuring that students learn the most up-to-date and relevant information. JHU is the leading institution in research, consistently conducting nearly twice the R&D of any other university, and has maintained this unparalleled position for decades.

INTERDISCIPLINARY APPROACH

The courses leverage JHU's strengths across various disciplines, including computer science, engineering, medicine, and public policy. They offer students a holistic understanding of AI's impact across different sectors and enable them to think broadly about its applications.



Experience the Johns Hopkins University advantage

Enroll in the Certificate Program in Applied Generative AI to drive AI innovation in your organization.

#6 NATIONAL UNIVERSITY
U.S. NEWS & WORLD REPORT

#13 BEST GLOBAL UNIVERSITY
U.S. NEWS & WORLD REPORT

WHO IS THIS PROGRAM FOR?

This program is designed for individuals looking to explore how Generative AI can solve real-world business problems.

Technology Professionals: Professionals who want to learn and apply Generative AI, enabling them to build and deploy AI-driven solutions at work or for personal projects using OpenAI and open-source LLMs.

Data Professionals: Data Analysts, Data Engineers, and Data Scientists seeking to enhance their ability to interpret and analyze data through Generative AI, uncovering deeper insights and expanding their expertise in text generation, image processing, and data analysis. They will also learn to deploy Generative AI solutions and strengthen their AI model development and data system maintenance abilities.

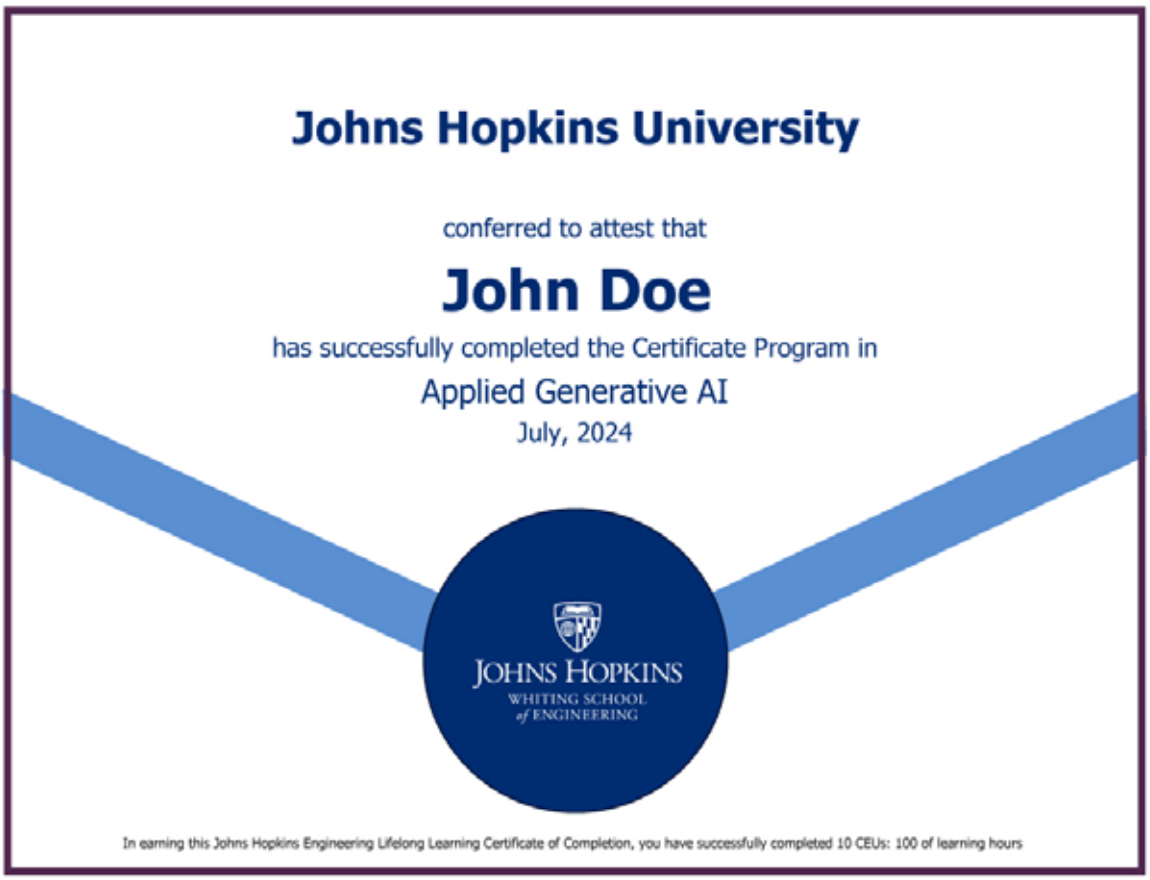
Technology Consultants and Technical Managers: Professionals who want to understand Generative AI, implement best practices, manage risks and ethical considerations, and guide technical teams to design and develop advanced AI solutions for their employers and clients.

STEM (Science, Technology, Engineering, and Mathematics) Graduates: Professionals who are graduates from science, technology, engineering, and mathematics fields wishing to upskill through hands-on training in Generative AI and become part of a cutting-edge industry with significant growth potential.

LEARNING OUTCOMES

- Understand the theoretical foundations of Generative AI and its applications.
- Apply Generative AI techniques to create text, image, and multimedia content.
- Implement best practices to mitigate potential risks in Generative AI solutions.
- Master the art of building intelligent AI agents using LangChain and ControlFlow to power real-world, agentic workflows and personalized automation.
- Develop and train Generative Models using contemporary Machine Learning frameworks.
- Evaluate the ethical implications of Generative AI.
- Critically analyze the impact of Generative AI on various industries and society as a whole.

CERTIFICATE OF COMPLETION



The image is for illustrative purposes only. The actual certificate may be subject to change at the discretion of Johns Hopkins University.

PROGRAM CURRICULUM

The curriculum, designed by the faculty of Johns Hopkins University, Great Learning, and leading industry practitioners, is taught by the best-in-class professors and practicing industry experts. The program aims to acquaint the learners with the skills needed to solve problems and deploy Generative AI solutions for various business applications.

PRE-WORK

This module provides all the necessary tools for your learning journey and establishes a solid foundation in three essential domains: Python programming, Introduction to AI and its applications, and Statistics.

MODULE 1: LEARNING PYTHON WITH GENERATIVE AI

This module provides a comprehensive introduction to Python programming fundamentals, focusing on Generative AI, and offers a solid understanding of how Large Language Models work.

WEEK 1: GENERATIVE AI LANDSCAPE

This week, you will learn the key concepts of Generative AI, how large language models (LLMs) function, what's under the hood, and why they behave the way they do. The week will conclude with an exploration of business applications of Generative AI across industries and functions, such as Marketing, Healthcare, and Productivity.

WEEK 2: PYTHON PROGRAMMING WITH GENERATIVE AI - PART 1

This week, you will learn how to use Generative AI models to generate code for simple Python-based applications, like a calculator or a database. You will also learn how to ask ChatGPT for lessons, generate code, interpret and debug code, and more.

WEEK 3: FOUNDATIONS OF AI

This week, you will learn the fundamentals of Machine Learning, which are essential for understanding how LLMs work at an intuitive level. Additionally, you will learn how to build ML classifiers using Generative AI and evaluate various Machine Learning models with the help of Generative AI.

WEEK 4: PYTHON PROGRAMMING WITH GENERATIVE AI - PART 2

This week, you will learn how to interact with Generative AI using various libraries, and you'll explore how to interface and work with different types of text and data modalities in Python. Building on the previous week, you will learn how to use Generative AI to store and manipulate text, read files, and clean text data in Python.

MODULE 2: GENERATIVE AI FOR BUSINESS PRODUCTIVITY

This module offers an opportunity to learn how to solve a variety of business problems using Generative AI. You will explore techniques such as text summarization, text classification, and text generation through prompting or Prompt Engineering with LLMs.

WEEK 5: NATURAL LANGUAGE PROCESSING

This week, you'll learn how to address fundamental text-based challenges, including sentiment analysis, topic modeling, and named entity recognition, through practical business case studies.

WEEK 6: TRANSFORMERS FOR LARGE LANGUAGE MODELS

This week, you will learn the foundational concept behind how Large Language Models (LLMs) work, specifically focusing on 'self-attention'. Additionally, you'll learn about the building block of LLMs, the 'Transformer'. Throughout the week, you will also learn how to solve text-based problems using Transformers.

WEEK 7: PROMPT ENGINEERING

This week, you will delve into the fundamentals of Prompt Engineering and prompting techniques to help you write effective prompts to obtain specific responses from an LLM for business use cases. You will learn about LLM training and development and model deployment. Additionally, you'll discover how to build useful workflows to automate problem-solving with LLMs using LangChain.

WEEK 8: CLASSIFICATION, CONTENT GENERATION AND SUMMARIZATION WITH GENERATIVE AI

This week, you will focus on addressing common issues in text processing, including summarization, content generation, and classification, in the context of Generative AI. Additionally, you'll learn how to assess the quality of these solutions using objective metrics and other LLMs.

WEEK 9: PROJECT 1 (GENAI SECRETARY-EMAIL SUMMARIZATION AND PRIORITY ASSISTANT)

Develop an AI-powered 'secretary' that assists users in managing emails more efficiently by highlighting the most urgent messages, summarizing email threads, and improving overall productivity. The project aims to leverage Generative AI models for classifying, prioritizing, and summarizing emails to provide concise actionable insights for users.

Tools and Concepts: Prompt Engineering, OpenAI, Classification, Summarization, LLM Evaluation

WEEK 10: LEARNING BREAK

MODULE 3: DESIGNING ADVANCED GENERATIVE AI WORKFLOWS

This module will focus on building and deploying advanced Generative AI solutions and agents using Retrieval Augmented Generation (RAG) and fine-tuned LLMs. You will learn to implement these technologies securely and responsibly for both private and public applications.

WEEK 11: SECURE AND RESPONSIBLE GENERATIVE AI SOLUTIONS

This week, you will be able to identify and mitigate bias and risk in AI systems, while also understanding and applying relevant laws and regulations for responsible usage of AI.

WEEK 12: DEVELOPING AGENTS WITH LANGCHAIN

This week, you will learn how to build Agentic Generative AI workflows with LangChain and create practical agents, such as Web and Database agents. Throughout this process, you will learn about LangChain library, AI agents, their types and workflows.

WEEK 13: RETRIEVAL AUGMENTED GENERATION (RAG) SEARCH

This week, you will understand the roles and differences between embeddings and tokenization in LLMs, learn the importance of Byte-Pair Encoding, gain insights into computing and applying sentence embeddings, explore how RAG improves response accuracy, and learn about the algorithms behind LLM embeddings and their impact on performance.

WEEK 14: ADVANCED RAG

This week, you will be able to differentiate between simplicity and depth in RAG implementations, fine-tune a basic RAG model, and evaluate RAGs effectively.

WEEK 15: FINE-TUNING AND CUSTOMIZATION OF GENERATIVE AI

This week, you'll learn how fine-tuning works, explore different fine-tuning methods, and see how to adjust an open-source LLM for real-world business uses.

WEEK 16: PROJECT 2 (AI FOR RESEARCH PROPOSAL AUTOMATION)

Build a Generative AI system to auto-draft NIH proposals with aligned ideas and high-quality content. The project aims to develop a Generative AI system that automates proposal drafting, identifies relevant past findings, and creates new project ideas aligned with the NOFO. This session is designed to reinforce key concepts, clarify doubts, and unblock progress. Project evaluation will be based on the quality of prompts and successful execution of the notebook.

Tools and Concepts: Prompt Engineering, OpenAI, RAG, Embeddings, Tokenization, Query Expansion, Summarization

TOOLS AND TECHNOLOGY

python™

Google Colab

gradio

RAG (Retreival Augmented Generation)

scikit learn

VS Code

Vector Database (Chroma / Pinecone)

ChatGPT

Whisper

OpenAI

LLaMA

Transformers

BERT

Quick Fine-Tuning Techniques

LangChain

SAMPLE CASE STUDIES

PREDICT LOAN DEFAULT CASES

Industry: Finance

Summary: Build a predictive model to identify high-risk loan applicants. Use structured evaluation to reduce subjectivity in approvals and flag potential defaults based on credit scores and loan types.

Tools and Concepts: Python, Model Building, Evaluation

SENTIMENT ANALYSIS ON SOCIAL MEDIA FOR NFL PLAYER DRAFT

Industry: Sports Analytics

Summary: Analyze Reddit discussions around NFL draft picks using NLP. Automate sentiment classification to extract insights for teams and analysts and support data-driven draft decisions.

Tools and Concepts: Tokenization, Embedding, Model Building

E-COMMERCE CHATBOT FOR ORDER MANAGEMENT

Industry: E-commerce

Summary: Improve customer support by deploying an AI-powered e-commerce chatbot built with responsible AI principles. Optimize response efficiency, reduce operational costs, and ensure fair, transparent, and secure user interactions.

Tools and Concepts: Open AI, Prompt Injection, Toxicity, Llama Guard, PII Protection, PII Masking, Gradio

SALESFORCE CRM Q&A ASSISTANT

Industry: Enterprise Software

Summary: Empower internal sales teams by implementing a RAG-powered CRM assistant. Optimize information retrieval to deliver instant, context-aware support, improving workflow efficiency and ensuring consistent, accurate responses.

Tools and Concepts: Embeddings, Tokenization, Retrieval Augmented Generation, RAGAs

MULTI-MODAL RAG FOR HEALTHCARE

Industry: Healthcare

Summary: Boost user engagement by launching a multimodal RAG-based AI system for personalized healthcare support. Optimize response relevance with text-and-image integration, enhancing trust and conversion in the wellness sector.

Tools and Concepts: Retriever with Metadata, Prompt Enhancement/Query Expansion, Multimodal RAG

Disclaimer: This is a tentative list and is subject to change.

PROGRAM FACULTY AND INDUSTRY MENTORS

FACULTY

Dr. Ian McCulloh leads the Johns Hopkins University Artificial Intelligence portfolio for Lifelong Learning. His research combines neuroscience, Artificial Intelligence, and human behavior to develop at-scale AI systems. McCulloh also served as an associate professor at Johns Hopkins and chief scientist at the Cyber Warfare Systems Group at the University's Applied Physics Laboratory. McCulloh retired as a Lieutenant Colonel from the U.S. Army after 20 years of service with expertise in social network analysis, special operations and improvised explosive device forensics. McCulloh holds a Ph.D. and Master's degrees from the Carnegie Mellon University School of Computer Science and the Florida State University, and a Bachelor's degree in Industrial Engineering from the University of Washington.

Dr. Pedro Rodriguez is a faculty member in the Johns Hopkins University Artificial Intelligence program and leads the Information Science branch at the Johns Hopkins Applied Physics Laboratory. He oversees a team of over 250 AI/ML researchers working on cutting-edge projects for the Department of Defense, the Intelligence Community, and other government agencies.

With over 20 years of experience, Dr. Rodriguez specializes in AI/ML algorithms for target detection, tracking, classification, and sensor fusion. He holds a Ph.D. in electrical engineering from the University of Maryland, Baltimore County, an M.S. in applied biomedical engineering from Johns Hopkins, and a B.S. in electrical engineering from the University of Puerto Rico.

Dr. Iain Cruickshank holds faculty positions in Computer Science at both Johns Hopkins University and Carnegie Mellon University, and serves as a Data Scientist and AI Engineer for the U.S. Army. He has held key roles at the Army's AI Integration Center and Army Cyber Institute, focusing on Machine Learning techniques to analyze online information environments and evaluate AI systems. His research has been applied to studying misinformation and disinformation campaigns, and he has published over 40 peer-reviewed papers.

Dr. Cruickshank holds a Ph.D. in Societal Computing from Carnegie Mellon, an M.S. in Operations Research from the University of Edinburgh, and a B.S. in Mathematics from West Point. He is a National Science Foundation Graduate Research Fellow and Rotary Ambassadorial Scholar.

Dr. Pavankumar Gurazada is a faculty member at Great Learning, specializing in business and AI, where he teaches AI and Machine Learning across various Master's programs. He holds a Ph.D. in Applied Machine Learning, with research focused on deep learning and MLOps. His work has been published in leading journals and conferences. In 2020, his book Marketing Analytics, published by Oxford University Press, became a popular textbook in post graduate programs. Dr. Gurazada also serves as a data science advisor and is a board member at Constems AI, a deep-tech startup developing computer vision systems for Industry 4.0.

INDUSTRY MENTORS

JEREMY SAMUELSON

Principal Data Scientist & ML Engineer at Equifax

BHASKARJIT SARMAH

Head RQA AI Labs at BlackRock

TANYA GLOZMAN

Applied Science, AI/ML at Apple

BRIDGET HUANG-GREGOR

Tech Lead, Engineering at Capital One

Note: This is an indicative list and is subject to change based on the availability of faculty and mentors.

ALUMNI SPEAK

Dustin Allen

AI Prompting Specialist | Custom GPT Designer, Ziplines Education

"Completing the Certificate Program in Applied Generative AI from Johns Hopkins University was both challenging and deeply rewarding. The structured learning path and dedicated mentorship helped me overcome initial anxiety and stay focused on my goals. With hands-on experience using tools like LangChain, vector databases, and RAG, the program built a strong foundation in applying AI to real-world challenges. My background in biotech and education made the content especially relevant across industries. The program ultimately empowered me to think creatively about AI's potential and how I can use it to make an impact."

Jackson Aaron

New Business Sales Executive, Dow Jones

"I joined the program to go beyond the buzz and truly understand how AI systems work. As someone in new business sales at Dow Jones, I've seen Generative AI's potential to transform workflows in finance, and I wanted to be hands-on with it.

The program offered weekly coding tasks, hands-on experience with LLM workflows, and real-world tools like LangChain and LangGraph. We built classifiers using noisy data, explored Prompt Engineering, autonomous agents, and more. The sessions on responsible AI were especially impactful, highlighting the importance of trust and fairness in technology. My final project; a Python-powered stock analysis bot, equipped me with practical skills that I now apply at my work.

This isn't a no-code program, it's designed for professionals eager to build, code, and learn by doing. If you're ready to get technical and push your learning further, it's absolutely worth the investment."

Rodrigo

Assistant Professor, Rollins School of Public Health at Emory University

"The academic team and Program Manager were exceptional, always available to answer questions, not just about lectures but also more advanced and related topics. The mentored sessions were a standout component and played a big role in the program's success. I would definitely recommend this program."

ADMISSION PROCESS

The admission process is conducted on a rolling basis and will be closed once the requisite number of candidates are enrolled.

APPLY
Fill out an online application form.

REVIEW
A panel from Great Learning will review your application to determine your fit for the program. They will evaluate your application on your academic performance, work experience, and motivation.

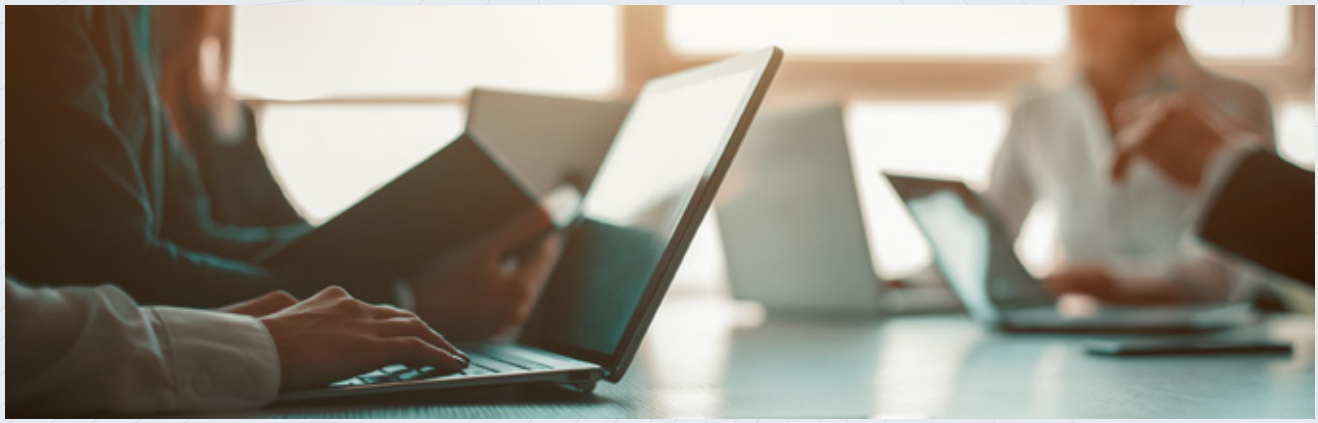
JOIN PROGRAM
After a final review, you will receive an offer for a seat in the upcoming cohort of the program.

PROGRAM FEE

$2,950

Please contact a Program Advisor from Great Learning for more information on offers, payment plans, and eligibility for financial assistance.

PROGRAM PARTNER



Great Learning
POWER AHEAD

Great Learning is a leading global ed-tech company for professional and higher education. It offers comprehensive, industry-relevant programs across various cutting-edge Technology, Data, and Business domains. These programs are developed in collaboration with the world's foremost academic institutions, such as MIT Professional Education, The University of Texas at Austin, Northwestern School of Professional Studies, Deakin University, Great Lakes Institute of Management, and more. They are constantly reimagined and revamped to address the dynamic needs of the industry. Offered in blended, classroom, and purely online modes, these programs are delivered with the help of expert mentors and highly qualified faculty. Great Learning is on a mission to enable transformative learning and career success in the digital economy, which has impacted 12 million+ learners from over 170 countries.

12 Mn+
Learners

170+
Countries

7700+
Industry Experts

CONTACT US

office-appl-genai-gl@jhu.edu
+1 410 584 3973

APPLY NOW